Lecture 8: The χ^2 Distribution

STAT 132

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1 The χ^2 (Chi-squared) Distribution

The chi-square (χ^2) distribution is important for many statistics applications. It is often used to analyze variance, test hypotheses, and evaluate goodness-of-fit. Here, we will introduce this distribution and look at some of its properties.

1.1 Definition

Definition 1.1. A continuous random variable X is said to follow a chi-square distribution with m degrees of freedom if it has the following pdf:

$$f(x|m) = \frac{1}{2^{m/2}\Gamma(m/2)} x^{(m/2)-1} e^{-x/2}, \quad x > 0.$$

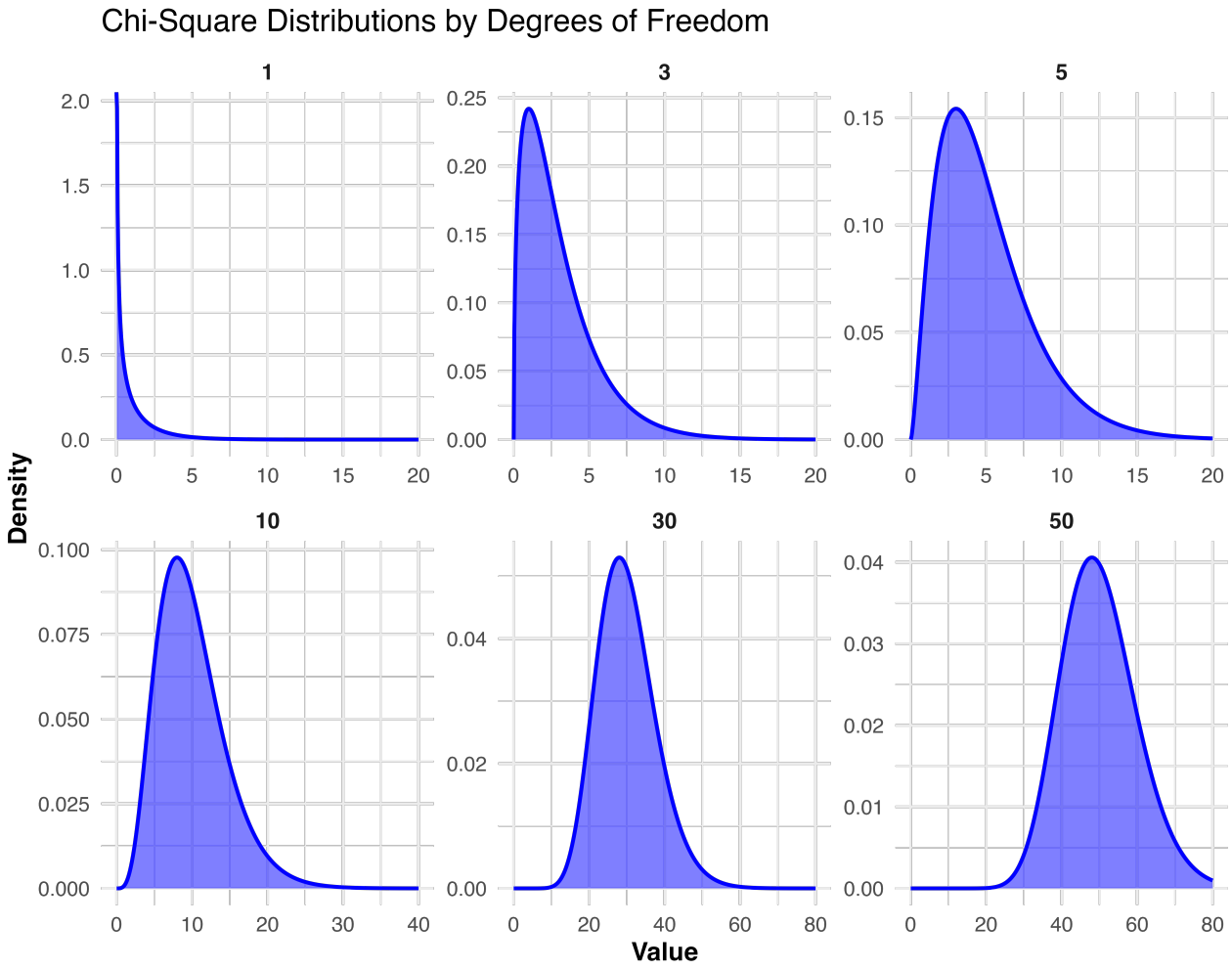
This is denoted as $X \sim \chi^2_m$.

Key Properties:

- **Mean:** $E(X) = m$.
- **Variance:** $\text{Var}(X) = 2m$.
- **Additivity:** If $X_1 \sim \chi^2_{m_1}$ and $X_2 \sim \chi^2_{m_2}$, and X_1 is independent of X_2, then $X_1 + X_2 \sim \chi^2_{m_1+m_2}$. Conversely, if $X_3 \sim \chi^2_{m_1+m_2}$ with $X_1 \sim \chi^2_{m_1}$, $X_2 \sim \chi^2_{m_2}$, and $X_3 = X_1 + X_2$, then X_1 and X_2 must be independent.
- **Origin:** The chi-square distribution is the sum of independent squared standard normal variables: $Z_1^2 + Z_2^2 + \cdots + Z_m^2 \sim \chi^2_m$.

Note: Although the parameter m (degrees of freedom) does not necessarily have to be an integer, in most cases it will be. We will see examples of this later.

1.2 Visualizing the χ^2 Distributions



The shape of the χ^2 distribution changes with degrees of freedom:

- For $m = 1$, the distribution is heavily skewed.
- As m increases, it becomes more symmetric, approaching a normal distribution.

1.3 Relation to Gamma Distribution

The pdf for the χ^2 distirbution should look very familiar. It is actually just a special case of the Gamma distribution where $\alpha = m/2$ and $\beta = 1/2$.

Theorem 1.1. *Suppose $X \sim \chi^2_m$. Then, for some constant c, the random variable $Y = cX$ will have a Gamma$(\frac{m}{2}, \frac{1}{2c})$ distirbution.*

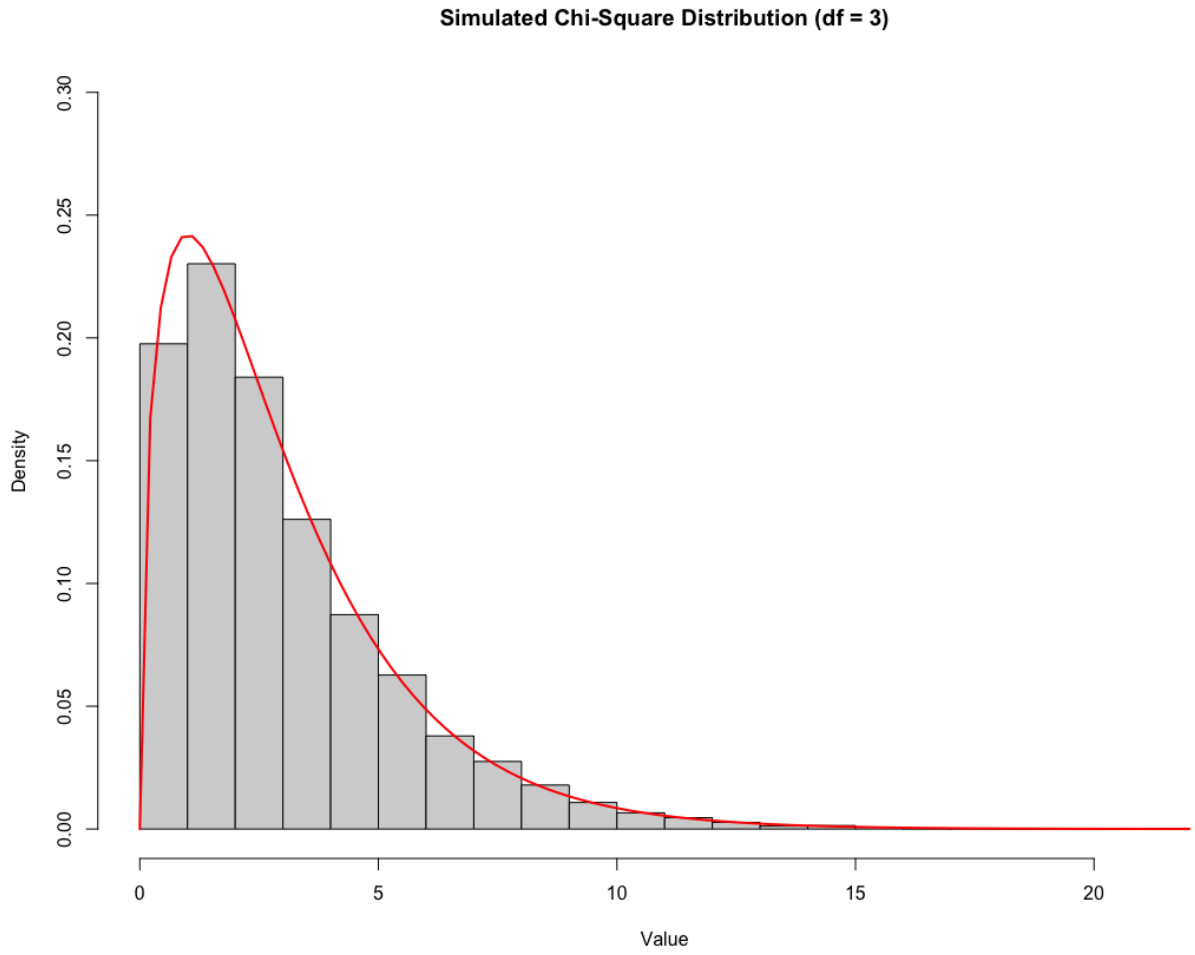
1.4 Relation to Normal Distribution

Theorem 1.2. *Suppose X follows a standard normal distribution. Then the random variable $Y = X^2$ will follow the χ^2_1 distribution.*

This theorem, along with the additivity property of χ^2 random variables, gives us a method to generate or sample χ^2 random variables, using samples from a standard normal distribution. More specifically, if the random variables $X_1, \ldots, X_m$ are i.i.d. standard normal, then $\sum_{i=1}^{m} X_i^2 = X_1^2 + \cdots + X_m^2$ will follow a χ^2_m distribution.

Example 1.1. Below, we simulate 10,000 instances of $Y = X_1^2 + X_2^2 + X_3^2$, where $X_i \overset{iid}{\sim} N(0,1)$, and compare the histogram to the theoretical χ^2_3 density.

```r
set.seed(123)
simulated_chi2 <- replicate(10000, sum(rnorm(3)^2))
hist(simulated_chi2, probability = TRUE,
     main = "Simulated Chi-Square Distribution (df = 3)",
     xlab = "Value", ylim = c(0, 0.3))
curve(dchisq(x, df = 3), add = TRUE, col = "red", lwd = 2)
legend("topright", legend = c("Simulated", "Theoretical"),
       col = c("black", "red"), lwd = 2)
```



♢

2 Applications

Example 2.1 (Variance Estimation)**.** Suppose that $X_1, \ldots, X_n$ form a random sample from a normal distribution with known mean μ and unknown variance σ^2. Then, the MLE for σ^2 is

$$\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Find the distribution of $n\widehat{\sigma^2}/\sigma^2$.

♢

Example 2.2. Consider the same setup as the previous example. Now, find the sampling distribution of $\widehat{\sigma^2}$.

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Example 2.3 (Sampling Distribution of Variance)**.** A manufacturer measures the weights $X_1, \ldots, X_n$ of $n = 10$ boxes of cereal. It is known that the cereal boxes are normally distributed and have a mean weight of $\mu = 12$ ounces. The manufacturer is interested in estimating the variance of this distribution. Accordingly, they calculated the MLE:

$$\widehat{\sigma^2} = \frac{1}{10}\sum_{i=1}^{10}(X_i - 12)^2.$$

Suppose that the true variance was actually $\sigma^2 = 0.25$. Find the probability that the MLE is between 0.2 and 0.4.

♢

3 χ^2 Table

Table of the χ^2 Distribution

If X has a χ^2 distribution with m degrees of freedom, this table gives the value of x such that $\Pr(X \leq x) = p$, the p quantile of X.

	p								
m	.005	.01	.025	.05	.10	.20	.25	.30	.40
1	.0000	.0002	.0010	.0039	.0158	.0642	.1015	.1484	.2750
2	.0100	.0201	.0506	.1026	.2107	.4463	.5754	.7133	1.022
3	.0717	.1148	.2158	.3518	.5844	1.005	1.213	1.424	1.869
4	.2070	.2971	.4844	.7107	1.064	1.649	1.923	2.195	2.753
5	.4117	.5543	.8312	1.145	1.610	2.343	2.675	3.000	3.655
6	.6757	.8721	1.237	1.635	2.204	3.070	3.455	3.828	4.570
7	.9893	1.239	1.690	2.167	2.833	3.822	4.255	4.671	5.493
8	1.344	1.647	2.180	2.732	3.490	4.594	5.071	5.527	6.423
9	1.735	2.088	2.700	3.325	4.168	5.380	5.899	6.393	7.357
10	2.156	2.558	3.247	3.940	4.865	6.179	6.737	7.267	8.295
11	2.603	3.053	3.816	4.575	5.578	6.989	7.584	8.148	9.237
12	3.074	3.571	4.404	5.226	6.304	7.807	8.438	9.034	10.18
13	3.565	4.107	5.009	5.892	7.042	8.634	9.299	9.926	11.13
14	4.075	4.660	5.629	6.571	7.790	9.467	10.17	10.82	12.08
15	4.601	5.229	6.262	7.261	8.547	10.31	11.04	11.72	13.03
16	5.142	5.812	6.908	7.962	9.312	11.15	11.91	12.62	13.98
17	5.697	6.408	7.564	8.672	10.09	12.00	12.79	13.53	14.94
18	6.265	7.015	8.231	9.390	10.86	12.86	13.68	14.43	15.89
19	6.844	7.633	8.907	10.12	11.65	13.72	14.56	15.35	16.85
20	7.434	8.260	9.591	10.85	12.44	14.58	15.45	16.27	17.81
21	8.034	8.897	10.28	11.59	13.24	15.44	16.34	17.18	18.77
22	8.643	9.542	10.98	12.34	14.04	16.31	17.24	18.10	19.73
23	9.260	10.20	11.69	13.09	14.85	17.19	18.14	19.02	20.69
24	9.886	10.86	12.40	13.85	15.66	18.06	19.04	19.94	21.65
25	10.52	11.52	13.12	14.61	16.47	18.94	19.94	20.87	22.62
30	13.79	14.95	16.79	18.49	20.60	23.36	24.48	25.51	27.44
40	20.71	22.16	24.43	26.51	29.05	32.34	33.66	34.87	36.16
50	27.99	29.71	32.36	34.76	37.69	41.45	42.94	44.31	46.86
60	35.53	37.48	40.48	43.19	46.46	50.64	52.29	53.81	56.62
70	43.27	45.44	48.76	51.74	55.33	59.90	61.70	63.35	66.40
80	51.17	53.54	57.15	60.39	64.28	69.21	71.14	72.92	76.19
90	59.20	61.75	65.65	69.13	73.29	78.56	80.62	82.51	85.99
100	67.33	70.06	74.22	77.93	82.86	87.95	90.13	92.13	95.81

"Table of the X2 Distribution" adapted in part from "A new table of percentage points of the chi-square distribution" by H. Leon Harter. From BIOMETRIKA, vol 51(1964), pp. 231–239.
"Table of the X2 Distribution" adapted in part from the BIOMETRIKA TABLES FOR STATISTICIANS, Vol. 1, 3rd ed., Cambridge University Press, © 1966, edited by E.S. Pearson and H.O. Hartley.

Table of the χ^2 Distribution *(continued)*

				p					
.50	.60	.70	.75	.80	.90	.95	.975	.99	.995
.4549	.7083	1.074	1.323	1.642	2.706	3.841	5.024	6.635	7.879
1.386	1.833	2.408	2.773	3.219	4.605	5.991	7.378	9.210	10.60
2.366	2.946	3.665	4.108	4.642	6.251	7.815	9.348	11.34	12.84
3.357	4.045	4.878	5.385	5.989	7.779	9.488	11.14	13.28	14.86
4.351	5.132	6.064	6.626	7.289	9.236	11.07	12.83	15.09	16.75
5.348	6.211	7.231	7.841	8.558	10.64	12.59	14.45	16.81	18.55
6.346	7.283	8.383	9.037	9.803	12.02	14.07	16.01	18.48	20.28
7.344	8.351	9.524	10.22	11.03	13.36	15.51	17.53	20.09	21.95
8.343	9.414	10.66	11.39	12.24	14.68	16.92	19.02	21.67	23.59
9.342	10.47	11.78	12.55	13.44	15.99	18.31	20.48	23.21	25.19
10.34	11.53	12.90	13.70	14.63	17.27	19.68	21.92	24.72	26.76
11.34	12.58	14.01	14.85	15.81	18.55	21.03	23.34	26.22	28.30
12.34	13.64	15.12	15.98	16.98	19.81	22.36	24.74	27.69	29.82
13.34	14.69	16.22	17.12	18.15	21.06	23.68	26.12	29.14	31.32
14.34	15.73	17.32	18.25	19.31	22.31	25.00	27.49	30.58	32.80
15.34	16.78	18.42	19.37	20.47	23.54	26.30	28.85	32.00	34.27
16.34	17.82	19.51	20.49	21.61	24.77	27.59	30.19	33.41	35.72
17.34	18.87	20.60	21.60	22.76	25.99	28.87	31.53	34.81	37.16
18.34	19.91	21.69	22.72	23.90	27.20	30.14	32.85	36.19	38.58
19.34	20.95	22.77	23.83	25.04	28.41	31.41	34.17	37.57	40.00
20.34	21.99	23.86	24.93	26.17	29.62	32.67	35.48	38.93	41.40
21.34	23.03	24.94	26.04	27.30	30.81	33.92	36.78	40.29	42.80
22.34	24.07	26.02	27.14	28.43	32.01	35.17	38.08	41.64	44.18
23.34	25.11	27.10	28.24	29.55	33.20	36.42	39.36	42.98	45.56
24.34	26.14	28.17	29.34	30.68	34.38	37.65	40.65	44.31	46.93
29.34	31.32	33.53	34.80	36.25	40.26	43.77	46.98	50.89	53.67
39.34	41.62	44.16	45.62	47.27	51.81	55.76	59.34	63.69	66.77
49.33	51.89	54.72	56.33	58.16	63.17	67.51	71.42	76.15	79.49
59.33	62.13	65.23	66.98	68.97	74.40	79.08	83.30	88.38	91.95
69.33	72.36	75.69	77.58	79.71	85.53	90.53	95.02	100.4	104.2
79.33	82.57	86.12	88.13	90.41	96.58	101.9	106.6	112.3	116.3
89.33	92.76	96.52	98.65	101.1	107.6	113.1	118.1	124.1	128.3
99.33	102.9	106.9	109.1	111.7	118.5	124.3	129.6	135.8	140.2